Binary search tree ordering

In a binary search tree with distinct keys, every key in a node's left subtree is smaller than the node key, and every key in its right subtree is larger. The same ordering rule applies recursively to each subtree.

Inorder traversal

Inorder traversal visits the left subtree, then the root, then the right subtree. Applied to a binary search tree with distinct keys, it lists the keys in ascending sorted order.

Preorder traversal

Preorder traversal visits the root before recursively visiting its left and right subtrees. The order is root, left, right. With explicit markers for missing children, this order can serialize a binary tree's structure.

Postorder traversal

Postorder traversal processes the left subtree, then the right subtree, and finally the root. This children-before-parent order is useful when freeing an entire tree, since children are handled before their parent.

Level order traversal

Level order visits all binary-tree nodes at one depth before moving to the next depth. A FIFO queue holds pending nodes. Remove a node from the front, visit it, then enqueue its existing left and right children.